LESSON PLAN

A/C MATERIAL (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	16	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Aerodynamic loading distribution, span wise and chord wise distribution, effects of sweep and distribution.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Aerodynamic loading distribution, span wise and chord wise distribution, effects of sweep and distribution.
5. Specific Objectives	15.1.1.1 Trainee will be able to explain aircraft structure, various structural elements and their details.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of wing structural elements and load paths; Safety: Ensure no loose items on desk; Instructor: Monitor trainee focus.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary structural elements of a wing?
(b) Define aerodynamic loading in the context of flight.
(c) Why is understanding load distribution critical for structural design?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of aerodynamic loading distribution • Span wise load distribution characteristics • Chord wise load distribution characteristics • Impact of wing sweep on load path • Effect of distribution on structural sizing	L & Ds	PPT, IB	15 min
(b)	• Analysis of span wise bending moments • Analysis of chord wise shear forces • Structural response to sweep variations • Load concentration at wing root • Optimization of spar placement	L & Ds	PPT, PDF	20 min
(c)	• Summary of sweep effects on loads • Comparison of straight vs swept wings • Implications for structural element design • Review of load distribution principles • Preparation for miscellaneous loading cases	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how wing sweep alters the span wise load distribution.
(b) Describe the relationship between chord wise distribution and spar design.
(c) Identify two key structural elements affected by aerodynamic loading.

12. Home Assignment and Follow Up	Prepare a brief note on how the principles of aerodynamic loading distribution discussed in this lesson apply to the miscellaneous loading cases covered in LP 07.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Miscellaneous loading cases, longitudinal cases, fuselage pressurization loads, flap and airbrake loads, spinning and power plants loads, fuel systems, freight loads, ground handling guided missile launcher cases.	Refer to the master syllabus reference list	LP 07

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min